

The codes in this repository are based on the Fernández (2012) wavelet approach and were originally developed in 2020, but uploaded to GitHub at a later date. The wavelet decomposition script uses MODWT with the LA8 filter. This project has been forked from the MODWT-MARS repository by 0zean and extended to include multiple wavelet correlation (MWC) analysis for 11 companies listed on the Tehran Stock Exchange.

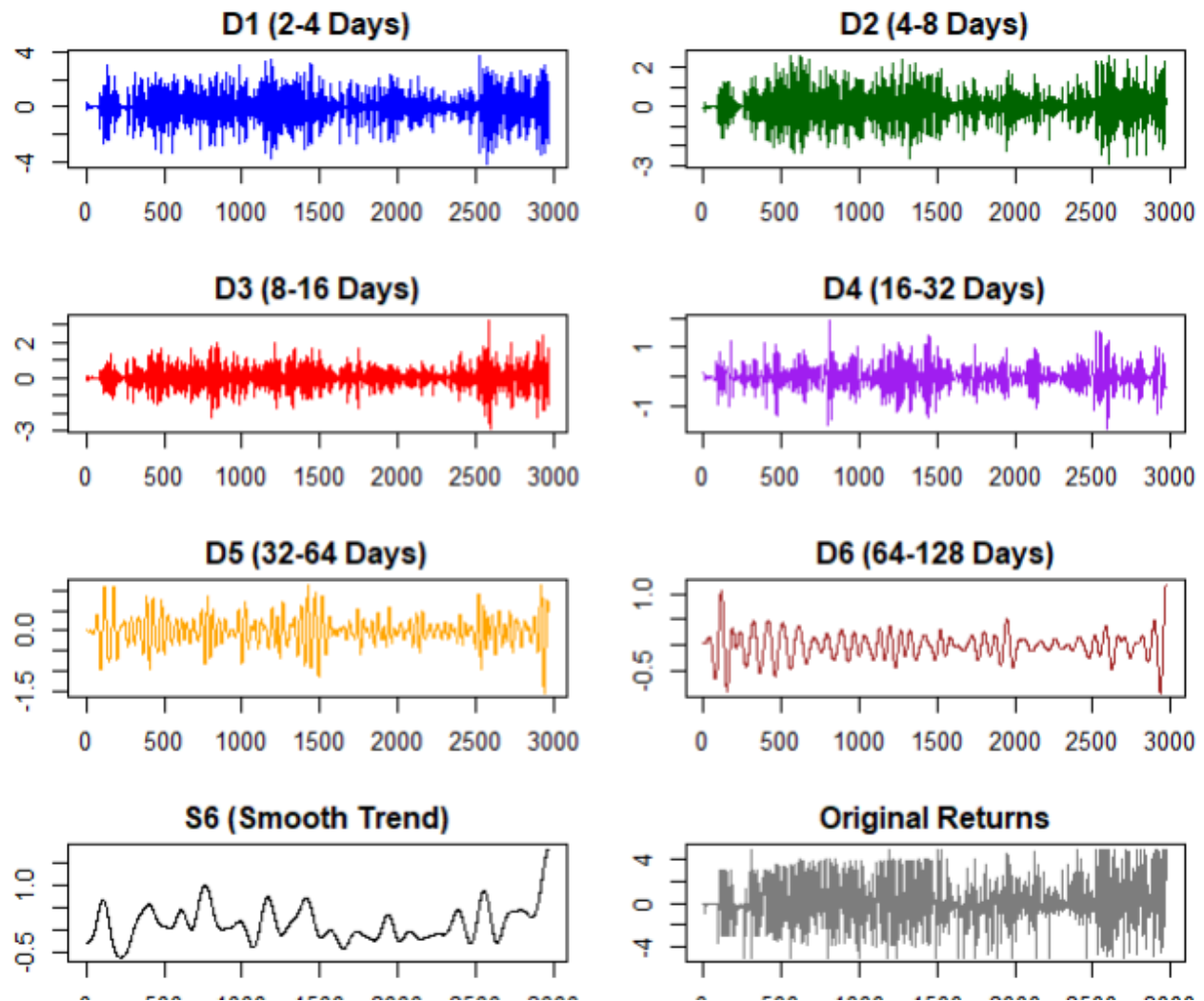


Figure 1: Wavelet Decomposition Output (D1–D6 & S6)

MODWT decomposition of the company returns into 6 detail levels (D1–D6) representing 2–128 day cycles, plus the smooth trend component (S6). (This code is forked from MODWT-MARS)

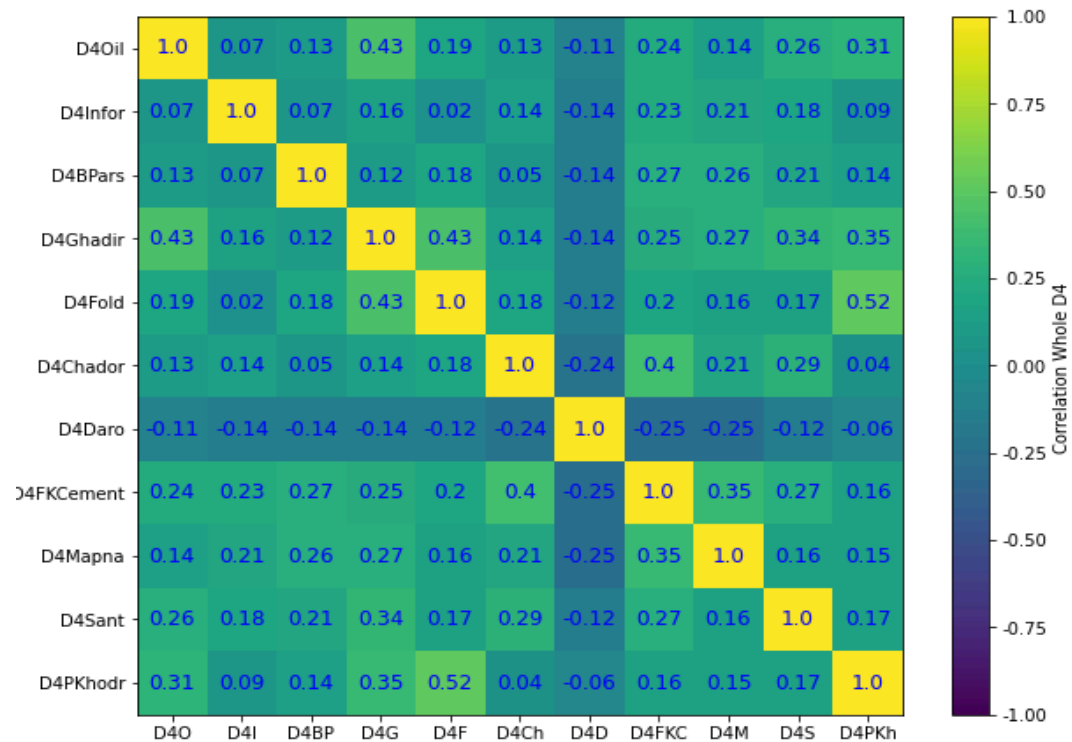


Figure 2: 11×11 Pairwise Correlation Matrix at D4 Level

Pairwise correlation coefficients between 11 companies at the D4 time scale (16–32 days), showing relationships ranging from -1.0 to 1.0.

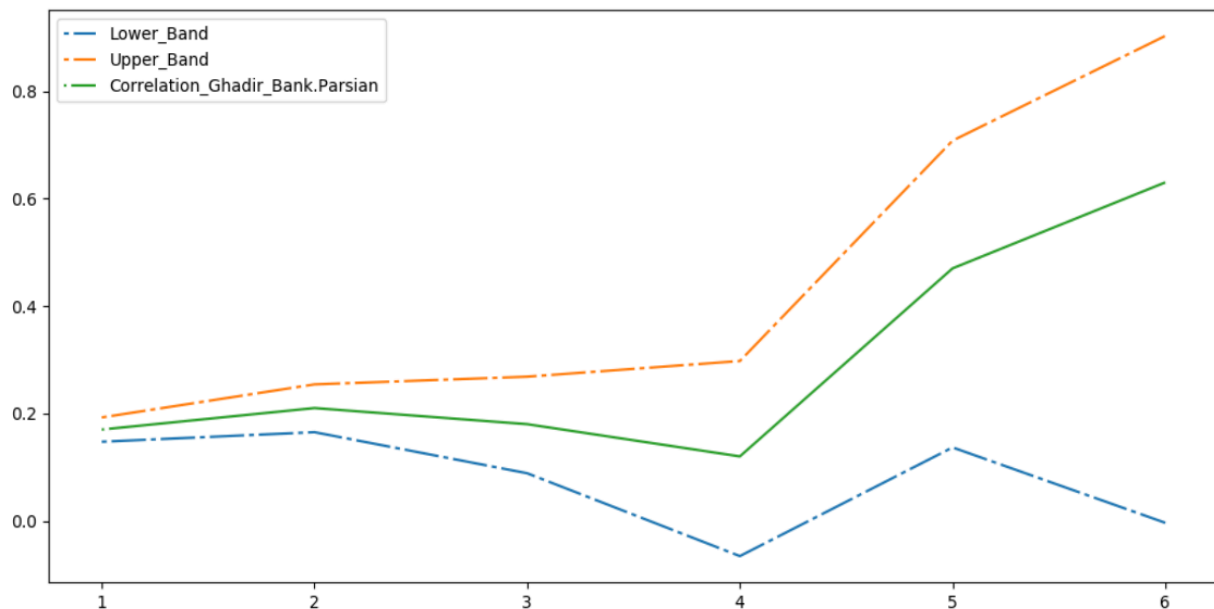


Figure 3: Correlation, Co-Movement, of Ghadir with Bank Parsian (D1–D6)

Correlation between Ghadir and Bank Parsian across 6 time scales, with upper and lower confidence bands.

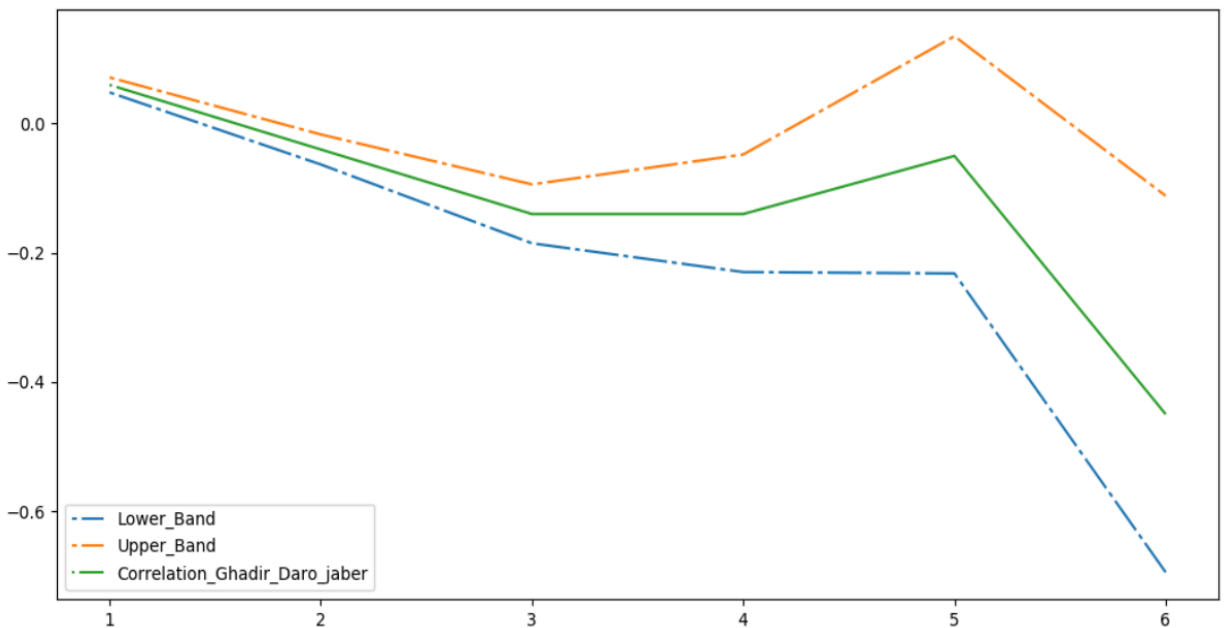
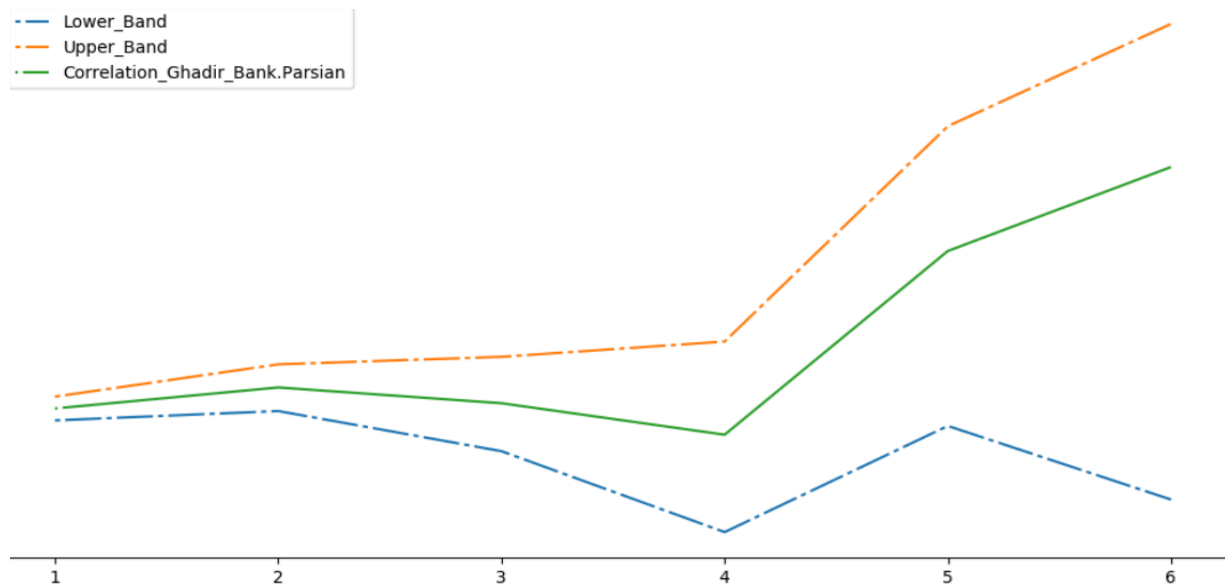


Figure 4: Correlation, Co-Movement, of Ghadir with Daro Jaber (D1-D6)
Correlation between Ghadir and Daro Jaber across 6 time scales, showing negative correlations at higher levels.



میزان هم حرکتی (همبستگی) سطح یک تا شش هولدینگ غدیر با بانک پارسیان شکل (۴-۱۶)

Figure 5: MWC Plot with Confidence Bands
Multiple Wavelet Correlation (MWC), Co-Movement, values for D1-D6 with 99.7% confidence bands, showing increasing correlation at longer time scales.